

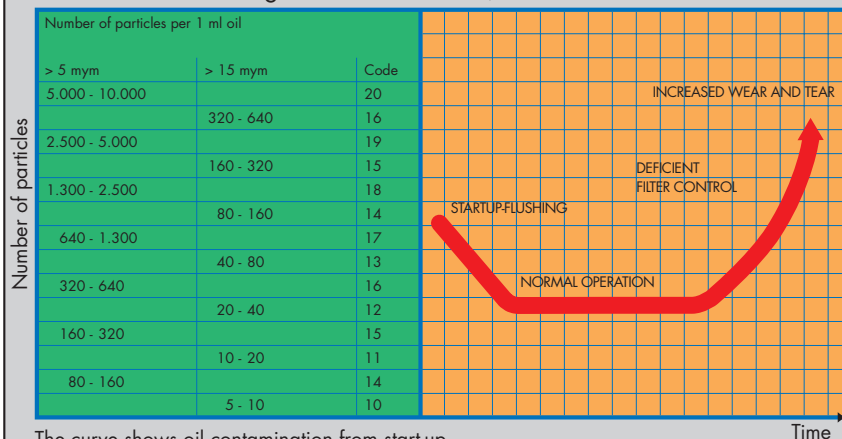
Frank Mohn Services AS

Service Bulletin

Condition monitoring of your cargo pumping system



Cleanliness according to ISO class 4406, recommended level code 16/12



The curve shows oil contamination from start-up through operational period, and the consequences of deficient monitoring of systems.

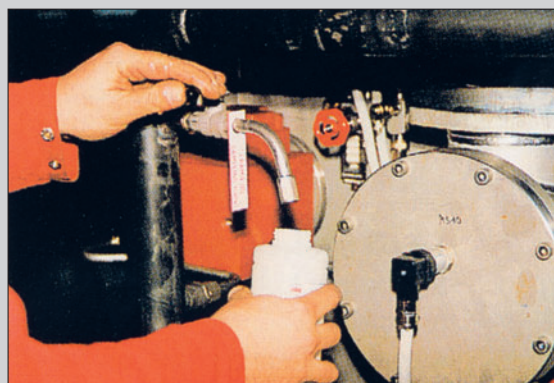
Oil sampling

Oil analysis

Filter
condition

Annual
inspection

Pre-docking
inspection



Same sampling point to be used each time.



Certified clean bottles



Particle counter Framo test laboratory



Karl Fisher water test equipment



Viscosity and density test equipment

Condition based maintenance has proven to be cost effective. Follow up the condition of the cargo pumping system will give improved lifetime of the components in the system.

To assist our customers with cost control and optimal operation, Framo can today offer the following assistance:

- Oil monitoring program.
- Annual inspections.
- Pre-docking inspections.

The primary advantages are:

- Secure optimal performance of your Framo cargo pumping system.
- Reduced maintenance cost.
- Increased lifetime of hydraulic components.
- Better possibility to plan the maintenance and to have spares available.
- Documentation of system condition.

Oil monitoring /Annual inspection/Pre-docking inspection:

The program will be managed by one of the Framo service stations which will:

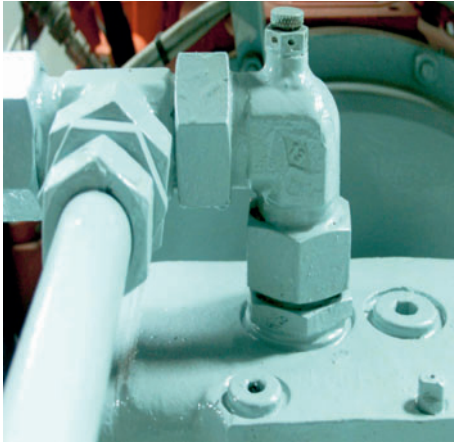
- Provide schedules for samples and inspections to ship and ship owner.
- Supply necessary sample bottles.
- Contact the ship when the next sample/inspection is due.
- Perform trend analysis of the different tests and samples and send this to ship and owner.
- Provide an annual report based on the available information from samples, inspection on board and consumption statistics.
- Pre-docking inspection prior to dockings.
- Proposal for actions required, if any.

The program comprise the following:

Level 1:

- Particle counting and water content test, quarterly.
The oil sample will be analysed at a Framo test laboratory according to ISO 4406.
 - Number and size of particles.
 - Water content analysis in accordance to Karl Fisher ASTM E-1064.

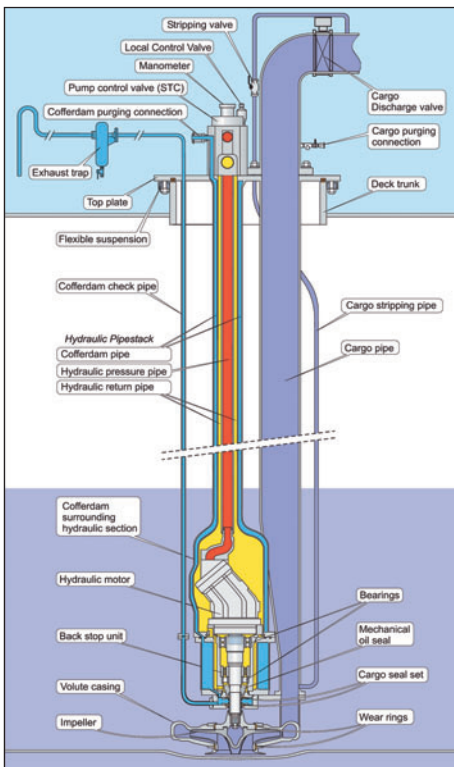
The program includes a total of 4 samples annually. On ships delivered after 1985, a separate sampling valve is a standard installation by Framo. On older vessels, a sample valve along with installation instructions will be provided as part of the agreement.



Manual wear indicator



Electronic wear indicator



The first year we will try to attend the ship to draw each sample, provided the ship is in one of the Framo ports. If this is not possible, we will ask the ship to send the samples to one of the Framo service stations. From the second year, the ship will be asked to send 3 of the samples to one of our Framo service stations world wide. The fourth sample is included in the level 2, full analysis.

Level 2:

- Full oil analysis, annually comprising the following:
 - Particle counting.
 - Water content, ppm (max 500 ppm).
 - Viscosity test at 40° C.
 - Total acid no.
 - Chlorides, ppm.
 - Phosphor, ppm.
 - Zinc, ppm.
 - Calcium, ppm.

Regarding the Level 2 sample we will attend the ship preferably in ports where Framo is represented. In addition to collecting the oil sample, we will also carry out an Annual Inspection of the Framo cargo pumping system.

Annual inspection:

The visual inspection will include the following:

- Hydraulic oil sampling and analysing.
- Visual inspection of hydraulic power unit.
- Inspection of manual wear indicators.
- Check for air in the system and deaerate if necessary.
- General inspection of the system for leakages, abnormal noise etc.
- Visual inspection and evaluation of hydraulic deck piping and valves, including quick coupling stations for portable pump connection.
- Inspection of cargo pump top plates, including FCV/STC valves.
- Evaluation of purging routines, status of the cargo pumps cofferdam.
- Visual inspection of Framo portable pump with accessories.
- Inspection of standard pumps and other Framo delivered equipment.
- Control of Framo toolbox/pressure testing kit.
- Inspection and evaluation of spare part inventory as well as stock location.
- Evaluation of planned maintenance records.
- Status report with recommendations.

Pre-docking inspection:

Prior to dry-docking, Framo will provide skilled engineers to attend your ship world wide in order to evaluate the condition of the entire Framo cargo pumping system. The pre-docking inspection is recommended carried out during a discharge operation or at a short voyage.

The inspection comprise the following:

- Test all cargo pumps against closed valve in order to evaluate condition of the pumps and control valves.
- Purge cofferdam on all cargo pumps.
- Test each hydraulic power pack.
- Test hydraulic control system, alarm and shutdown functions.
- Visual inspection of hydraulic piping and service valves.
- Inspection and test of portable pump if possible.
- Test and evaluation of ballast pumps and tank cleaning pumps.
- Inspection of spare part inventory and Framo toolbox.
- Following the survey, a detailed inspection report with repair recommendations will be sent to the shipowner/management company. Upon request, a quotation for spare parts will also be given. This will help the ship's superintendent in preparing the dry-docking activities on the Framo system, and at the same time keep the maintenance cost down.
- In the year when pre-docking inspection is carried out, it is not deemed necessary to also carry out an annual inspection.
- Pre-docking inspection can also be combined with on board training.

For any further information regarding our services, please do not hesitate to contact any of the Framo service stations.



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